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Estimates of egg deposition and effects of lake trout (*Salvelinus namaycush*) egg predators in three regions of the Great Lakes

Jory L. Jonas, Randall M. Claramunt, John D. Fitzsimons, J. Ellen Marsden, and Brian J. Ellrott

Abstract: Excessive mortality from spawning through the first year of life is likely responsible for recruitment failures in many lake trout (*Salvelinus namaycush*) populations. We evaluated relationships between interstitial predator species and lake trout egg deposition in three regions of the Great Lakes. The introduced crayfish *Orconectes rusticus* dominated the nearshore zone in northern Lake Michigan, whereas other sites in Lake Michigan and Parry Sound were dominated by native *Orconectes propinquus*. In Lake Champlain, sculpin (*Cottus* spp.) were the most common interstitial predator, yet densities were comparable with those in Lake Michigan. Predator densities in Lake Michigan (mean $\pm$ standard error = 15.5 ± 0.4 predators·m⁻²) were twice those in Lake Champlain or Parry Sound (6 ± 1 predators·m⁻²). Densities of eggs were higher in Lake Champlain (2994.1 ± 398.3 eggs·m⁻²) and Parry Sound (454.7 ± 36.3) than in Lake Michigan (7.5 ± 1.3 ; Kruskal–Wallis, $P \leq 0.05$). Ratios of eggs to predator provide simple measures of the effectiveness of lake trout rehabilitation efforts. The average ratio calculated from 10 successful spawning locations is 190 eggs·predator⁻¹·m⁻². Ratios may help establish target egg densities for restoration efforts and the concentration of spawning adults required to allow survival to the fry stage.

Résumé : L'insuccès du recrutement chez plusieurs populations de touladis (*Salvelinus namaycush*) s'explique par une mortalité excessive après la fraye et au cours de la première année de vie. Nous avons évalué les relations entre les espèces de prédateurs dans la zone interstitielle et la ponte d'oeufs de touladis dans trois régions des Grands Lacs. L'écrevisse introduite *Orconectes rusticus* domine la zone proche du rivage dans le nord du lac Michigan, alors que l'espèce indigène *Orconectes propinquus* prédomine dans les autres sites du lac Michigan et à Parry Sound. Au lac Champlain, les chabots (*Cottus* spp.) sont les prédateurs les plus communs dans la zone interstitielle, mais les densités sont comparables à celles du lac Michigan. Les densités de prédateurs au lac Michigan (moyenne $\pm$ écart type = $15,5 \pm 0,4$ prédateurs·m⁻²) sont deux fois plus élevées que celles au lac Champlain ou à Parry Sound (6 ± 1 prédateurs·m⁻²). Les densités d'oeufs sont plus élevées au lac Champlain ($2994,1 \pm 398,3$ oeufs·m⁻²) et à Parry Sound ($454,7 \pm 36,3$) qu'au lac Michigan ($7,5 \pm 1,3$; test de Kruskal–Wallis, $P \leq 0,05$). Le nombre d'oeufs par prédateur fournit une mesure simple de l'efficacité des efforts de réhabilitation du touladi. Le rapport moyen calculé dans 10 sites de fraye « réussie » est de 190 oeufs·prédateur⁻¹·m⁻². Ces rapports peuvent servir à fixer les densités d'oeufs à atteindre dans les efforts de restauration et à déterminer les concentrations d'adultes reproducteurs nécessaires pour permettre la survie jusqu'au stage alevin.

[Traduit par la Rédaction]

Introduction

Current research suggests that excessive mortality between the time of spawning in the fall and the first year of life is likely responsible for recruitment failures in lake trout (*Salvelinus namaycush*) populations throughout most of the Great Lakes (Jones et al. 1995; Savino et al. 1999). By the

1960s lake trout in all of the lakes except Lake Superior and the Parry Sound portion of Lake Huron were extirpated due to overfishing and sea lamprey (*Petromyzon marinus*) predation. With exceptions for Lake Superior and Parry Sound, stocking during the last four decades has generally failed to produce substantial recruitment. In Lake Michigan, we know that lake trout are not recruiting to the fishery, yet have

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some evidence of reproductive efforts (eggs, fry, and rarely young-of-the-year). The extent of these efforts in relation to other more successful systems has not been evaluated.

Relative abundance and age composition estimates are often the only parameters evaluated when determining the status of lake trout populations. A comprehensive synthesis of gillnet catch rates and their implied relation to adult abundance established the metric (50 lake trout $\cdot$ 305 m⁻¹ of gillnet; range, 17–135) that many managers currently target to establish spawner abundance levels capable of producing measurable recruitment and self-sustaining stocks (Selgeby et al. 1995). As a result, investigations into the spawning potential of lake trout have primarily focused on measurements of the abundance of spawning adults (Holey et al. 1995; Selgeby et al. 1995) on specific reefs with only occasional estimates of egg deposition (Peck 1986; Perkins and Krueger 1995; Chotkowski and Marsden 1997, 1999). Evaluations of the influence of mortality factors in the early life stages of lake trout life history have been nearly non-existent, and the relationships to spawning stock size and habitat use have not been established (Horns and Magnuson 1981; Kelso 1995).

To truly characterize mechanisms behind recruitment failures and evaluate reproductive potential, estimates of adult abundance during spawning are not sufficient. For measurable natural reproduction to occur, enough eggs must be deposited to survive interstitial predation and develop over the winter into emergent fry, after which they must escape the gauntlet of piscivorous predators offshore (Gunn 1995). The inability to reach any of these stages results in recruitment failure. In Lake Ontario, Fitzsimons et al. (2002) measured predator abundance levels that were high enough to cause mortality of all eggs when deposition rates were low. When egg deposition increased, predator impacts were reduced. Our current knowledge of predation losses in the early life history of lake trout is based almost entirely on laboratory studies and modeling efforts (Jones et al. 1995; Savino et al. 1999); these indicate severe consequences to recruitment but have yet to be validated in the field.

The potential effects of native and non-native interstitial predators on lake trout recruitment are strongly dependent on the density of naturally deposited eggs, the species composition, density and size of predators, and their underlying relationships (Jones et al. 1995; Savino et al. 1999). Each species of predator has a finite ability to consume eggs that is likely regulated by their size (Evans et al. 1987; Post and Lee 1996) and water temperature (Thornton and Lessem 1978; Carline 1987; Hill and Magnuson 1990). The relative importance of egg predation may be even greater now, with the recent invasion of rusty crayfish (*Orconectes rusticus*) and round gobies (*Neogobius melanostomus*), two species known to consume lake trout eggs at higher rates than native crayfish and sculpins (Capelli 1982; Chotkowski and Marsden 1999; Clapp et al. 2001). Moreover, both round gobies and rusty crayfish are documented competitors for food resources and habitat, and are highly fecund when compared with native species (Capelli 1982; Olsen et al. 1991; Chotkowski and Marsden 1999). Rusty crayfish are known to interbreed with *Orconectes propinquus* (Capelli and Capelli 1980). Site-specific estimates of predator species abundance, species composition, and size distributions would better define the effects of predation on lake trout recruitment success.

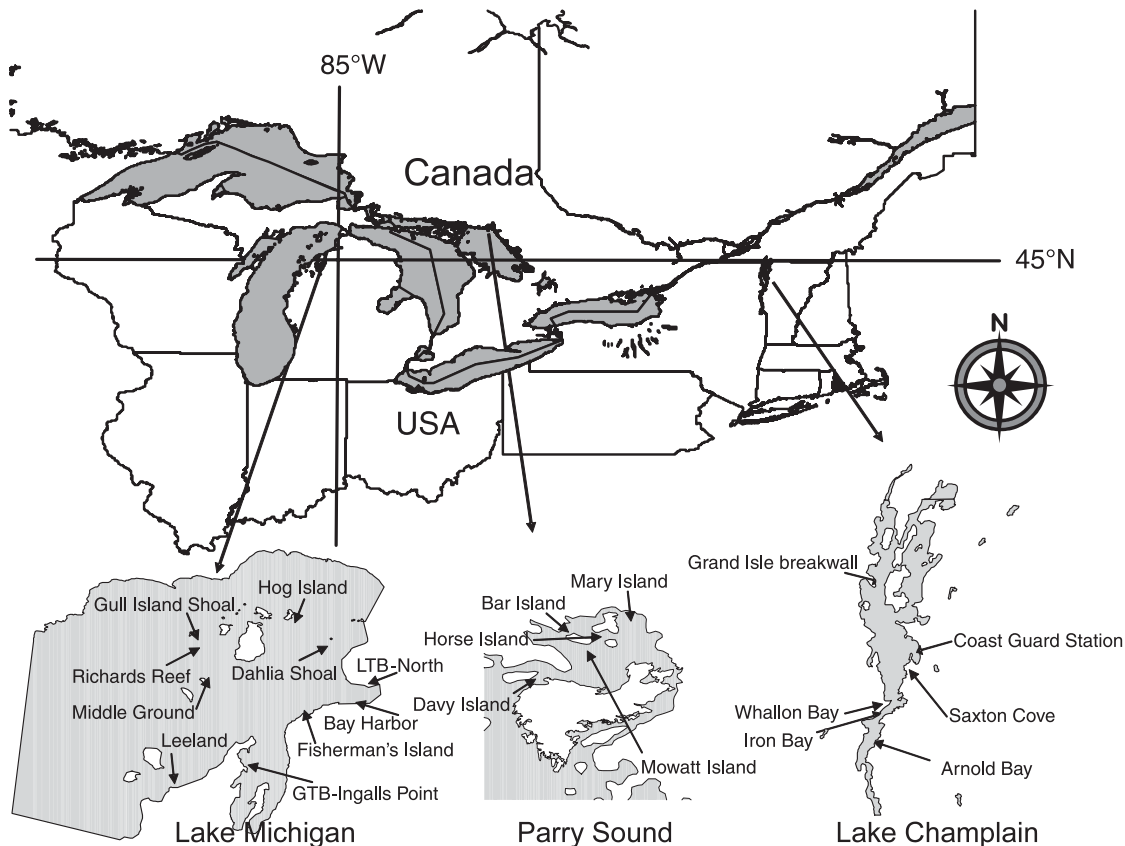
The goals of our study were as follows: (i) determine the number of naturally deposited lake trout eggs surviving to overwinter on several spawning reefs in Lake Michigan, Parry Sound (Lake Huron), and Lake Champlain; (ii) determine the density, species composition, and size ranges of interstitial predators on these reefs; and (iii) evaluate egg and predator densities to determine target egg-to-predator ratios that will lead to emergence of lake trout fry, a precursor to measurable recruitment.

Methods

We measured lake trout egg densities after exposure to interstitial predation and prior to the overwinter period in three diverse large lake systems (Lake Michigan, Parry Sound (Lake Huron), and Lake Champlain; Fig. 1). Systems were selected because they provide extensive habitat for lake trout reproduction, and are focal areas for lake trout restoration efforts (Holey et al. 1995; Reid et al. 2001; Ellrott and Marsden 2004). Within a given system, spawning sites were identified through reconnaissance diving surveys of habitat, past estimates of lake trout egg deposition, and existing knowledge of sites known to attract spawning lake trout in the fall. Once selected, divers surveyed entire reefs to find areas with substrate, slope, and depth characteristics appropriate for attracting spawning lake trout (Fitzsimons 1994; Marsden et al. 1995).

In Lake Michigan, no remnant stocks of lake trout remain, target stocking levels have never been achieved, lamprey populations (while significantly lower than during the pre-control period) continue to kill significant numbers of lake trout, and in some areas, commercial and recreational exploitation rates have remained high (Holey et al. 1995; Bronte et al. 2003b). When measured, egg deposition has been low or absent (Marsden 1994; Holley et al. 1995; Marsden and Janssen 1997), and the relationships between egg density and interstitial predators have not been evaluated. Most of the potential lake trout spawning habitat is found in the northern part of the lake, which is the focal area of our investigations (Dawson et al. 1997). Historically, lake trout co-existed with an interstitial community that included native crayfish (*Orconectes propinquus* and *Orconectes virilis*) and sculpin (*Cottus bairdi* and *Cottus cognatus*). However, the nearshore reef communities in Lake Michigan have recently been dramatically altered by the invasion of exotic egg predators (i.e., rusty crayfish and round gobies). For analyses, we split northern Lake Michigan into four regions: a protected bay with large populations of native predators (Ingalls Point in Grand Traverse Bay), a nearshore region with small numbers of native predators (Leeland), three nearshore locations with large populations of exotic predators, and five offshore locations with native predator communities (Fig. 1). Much of the potential lake trout spawning habitat in northern Lake Michigan is represented by the on-shore exotic predator and offshore native predator habitats. The offshore sites were Dahlia Shoal, Middle Ground, Hog Island Reef, Gull Island Shoal, and Richards Reef. Of these locations, only Gull Island Shoal and Richards Reef had been stocked in the past 15 years. The nearshore locations were Bay Harbor (Little Traverse Bay – South), Fisherman's Island, and a man-made crib structure (Little Traverse Bay –

Fig. 1. Map of the northeastern United States and southern sections of the Canadian provinces Ontario and Québec indicating regions evaluated and study-site locations within regions.



North). Of the nearshore locations, Bay Harbor (Little Traverse Bay – South), Fisherman's Island, and Leeland have been stocked in the past 15 years.

In contrast with Lake Michigan, lake trout populations in Parry Sound (Lake Huron) are now self-sustaining, and managers have declared success in rehabilitation efforts (Reid et al. 2001). Stocking was discontinued in 1998, remnant stocks of naturally reproducing lake trout remain, and older (>8 years) spawning lake trout are abundant in the population (Reid et al. 2001). This region serves as an important reference point for evaluating successful levels of egg deposition in the presence of native egg predators. In Parry Sound, adult fish are likely depositing sufficient numbers of eggs on spawning reefs to overcome predation, yet the dynamics of egg deposition and egg predation are not well documented. Five nearshore sites (Davy, Bar, Mowatt, Horse, and Mary islands), identified as active lake trout spawning reefs in previous investigations, were selected for this project (J.D. Fitzsimons, personal communication; Fig. 1).

Lake Champlain represented an intermediate case, in which predators are mainly native species, there are no remnant stocks of lake trout, and only 5%–10% of adult lake trout are naturally produced (Ellrott and Marsden 2004). Key factors implicated in lake trout recruitment failures throughout the Great Lakes (e.g., alewives, Krueger et al. 1995; Fitzsimons et al. 1999; high contaminant burdens, Fitzsimons 1995b; and excessive overexploitation, Stanley et al. 1987; Holey et al. 1995) are absent in Lake Champlain, creating a unique opportunity to evaluate the relative importance of egg predation

in limiting recruitment. We selected three nearshore sites based on reconnaissance diving surveys and anecdotal stories from local fisheries managers and anglers: Grand Isle, Arnold Bay, and Whallon Bay. An additional three sites were identified based on habitat characteristics alone (Fitzsimons 1994; Marsden et al. 1995): Saxton Cove, Iron Bay, and Coast Guard Station (Fig. 1).

We used egg bags to collect lake trout eggs and their predators in the fall (Perkins and Krueger 1994, 1995; Fitzsimons 1995a). Each egg bag was constructed from a 0.5 m deep nylon mesh material (3-mm mesh) attached to a polyvinyl chloride ring (0.35 m in diameter or 0.96 m² in area). Divers deployed nets by excavating a 40 to 50 cm deep hole, placing the net in the hole, and back filling with the removed substrate. Egg bags ($N = 60$) were buried at each location prior to the spawning season (August–September). The bags were placed in pairs along either side of a 30 m long marked line at 1-m intervals. The bags were positioned at the top of the drop-off at each location. Bags were retrieved after the spawning season (late October – early November). Egg densities reported in this investigation represent the number of eggs surviving predation approximately 4 weeks past peak spawning (late November – early December). Divers would carefully remove the substrate from inside the bag, while preventing interstitial predators from escaping. Predator escapes were rare, readily observed, and recorded by the divers. Recovered bags were brought to the lab where all eggs and predators were counted. We calculated the average number of eggs per square metre and the associated

variance (standard error (SE)) for each site and lake. Interstitial predators were identified, counted, weighed (g), and measured (mm). Total lengths were recorded for all fish species, and carapace lengths were recorded for all crayfish species. For each site, we estimated the density per square metre of individual predator species and of all predators combined. We only included individuals that were large enough to prey on lake trout eggs. Crayfish with carapace lengths greater than 19 mm and sculpin and round gobies greater than 42 mm total length were considered viable lake trout egg predators (Chotkowski and Marsden 1999; Fitzsimons et al. 2002).

Egg abundance, predator abundance, and predator length data did not meet the assumptions of normality or homogeneity of variance. Log and square root transformations did not normalize the data. Therefore we use the non-parametric Kruskal-Wallis (K-W) one-way analysis of variance by ranks to evaluate differences in egg and predator density among sites and regions. After determining significance in the overall test, multiple comparisons within groups were completed by calculating the critical difference value using z at $\alpha = 0.05$ for $k(k - 1)$ comparisons (Siegel and Castellan 1988). We also calculated the average length of each predator species by site and lake, and estimated the variance (SE).

To establish a measure of the interaction between egg deposition and predator abundance and to determine the relationship to fry emergence in the spring, we estimated the number of lake trout eggs deposited per predator within a square metre area at each location. Estimates of eggs per predator were determined for sites with and without fry emergence in the spring.

Emergence in the spring was measured using 10–30 emergent fry traps at each site (J.E. Marsden, B.J. Ellrott, R.M. Claramunt, J.L. Jonas, and J.D. Fitzsimons, unpublished data). Traps were checked weekly and deployed for 6–10 weeks beginning in mid-April, when ice no longer prohibited access to sites. The emergence of fry provided a measure of success in surviving predation and the overwinter period.

Results

Predator species composition, density, and size

Systems that were successful in producing measurable fry emergence tended to have lower densities of interstitial predators. Lake Champlain and Parry Sound both had low predator densities, averaging 6 predators·m⁻² (range, 2–11 m⁻²; K-W, $P > 0.05$), which was less than one-half that of Lake Michigan at 15.5 ± 0.4 predators·m⁻² (range, 8–27 m⁻²; K-W, $P \leq 0.05$; Table 1).

We observed eight native and two exotic species in egg collection bags from sites in Lake Michigan. Potential egg predators included four fish species (mottled and slimy sculpin, common white suckers (*Catostomus commersoni*), and longnose suckers (*Catostomus catostomus*)) and two crayfish (*O. propinquus* and *O. virilis*); we also collected johnny darter (*Etheostoma nigrum*), longnose dace (*Rhinichthys cataractae*), smallmouth bass (*Micropterus dolomieu*), and mudpuppies (*Necturus maculosus*). Darters, dace, and minnows are not considered viable predators: they are gape limited and do not consume lake trout eggs that are relatively

large (5–5.6 mm diameter; Auer 1982). Young-of-the-year smallmouth bass were evaluated in the lab and did not consume lake trout eggs (J.D. Fitzsimons, G. Fodor, and B. Williston, unpublished data). The two exotic species, both egg predators, were rusty crayfish and round goby. A large number of longnose dace were collected at one nearshore location (Leeland); though not egg predators, they were placed in the “other” category to illustrate the unique attributes of this location (Fig. 2).

Egg predators accounted for 97% of the interstitial species collected in Lake Michigan and the abundance of predators did not differ between offshore (14.4 ± 0.6 m⁻²) and nearshore (16.0 ± 0.6 m⁻²) locations (K-W, $P = 0.08$). Crayfish and sculpin were most abundant, crayfish (9.1 ± 0.3 m⁻²) being significantly more abundant than sculpin (6.1 ± 0.3 m⁻²; K-W, $P < 0.001$; Fig. 2). Rusty crayfish and the round goby were found only at nearshore locations, and rusty crayfish (7.9 ± 0.5 m⁻²) were more abundant than round gobies (0.8 ± 0.2 m⁻²; K-W, $P < 0.001$; Fig. 2). Average densities of the three species of crayfish differed (K-W, $P < 0.05$); *O. rusticus* was most abundant (7.9 ± 0.5 m⁻²), followed by *O. propinquus* (4.2 ± 0.2 m⁻²) and *O. virilis* (2.1 ± 0.2 m⁻²; Fig. 3). The average density of round gobies was lower (0.8 ± 0.2 m⁻²) than that of sculpin (5.8 ± 0.3 m⁻²; K-W, $P < 0.001$). However, sculpin densities were lower at sites with round gobies (2.5 ± 0.2 m⁻²) when compared with locations where they were absent (7.7 ± 0.4 m⁻²; K-W, $P < 0.001$). Similarly, *O. propinquus* densities were much lower (0.8 ± 0.2 m⁻²) at nearshore sites, where they co-existed with rusty crayfish, when compared with sites where they were absent (7.1 ± 1.6 m⁻²; K-W, $P < 0.001$; Fig. 3).

In Parry Sound we observed eight native species in egg collection bags: five fish (two predators, rock bass (*Ambloplites rupestris*) and sculpin (slimy and mottled), and three non-predators, bluntnose minnow (*Pimephales notatus*), largemouth bass (*Micropterus salmoides*), and smallmouth bass) and three crayfish (*O. propinquus*, *O. virilis*, and *Cambarus robustus*); no exotic species were observed. Crayfish were most abundant in Parry Sound, accounting for over 71% of the interstitial species collected. *Orconectes propinquus* was more abundant (3.9 ± 0.4 m⁻²; K-W, $P < 0.05$) than *Cambarus robustus* (0.7 ± 0.2 m⁻²) and *O. virilis* (0.2 ± 0.1 m⁻²; K-W, $P > 0.05$; Fig. 3). With the exception of the nearshore zone in northern Lake Michigan, which was dominated by *O. rusticus*, sites in Parry Sound and Lake Michigan were primarily occupied by *O. propinquus* (Fig. 3). Fish accounted for less than 30% of the species collected, and only 50% of these were potential egg predators (Fig. 2). In total, potential egg predators (both fish and crayfish) accounted for 86% of the species collected in egg bags in Parry Sound. Predator abundance levels did not differ among sites in Parry Sound (average = 5.5 ± 0.5 predators·m⁻²; K-W, $P = 0.057$; Table 1).

Within Lake Champlain, we observed seven native interstitial species in egg bags, including six fish (two predators, rock bass and sculpin (slimy and mottled), and four non-predators, johnny darter, banded killifish (*Fundulus diaphanous*), smallmouth bass, and log perch (*Percina caprodes*)) and one crayfish (*Orconectes limosus*). Egg predators accounted for 91% of the interstitial species collected in egg bags. Unlike Lake Michigan and Parry Sound, sculpin ($4.9 \pm$

Table 1. Number of egg nets evaluated (*N*), average number of lake trout (*Salvelinus namaycush*) eggs deposited, abundance of all predators large enough to consume lake trout eggs, and the ratio of eggs to predator per square metre in Parry Sound and Lakes Michigan and Champlain.

Site	<i>N</i>	Eggs·m ⁻² ± SE	Total predators·m ⁻² ± SE	Egg-to-predator ratio	Fry present
Parry Sound	379	454.7±36.3	5.5±0.5	82.1	y
Bar Island	89	808.2±90.9 x	3.6±0.8 a	222.0	y
Davy Island	89	723.7±100.3 x	4.9±0.8 a	147.5	y
Horse Island	88	214.0±25.1 y	5.6±1.0 a	38.2	y
Mary Island	24	29.9±9.1 z	7.0±2.2	4.3	n
Mowatt Island	89	184.5±28.7 y	7.6±1.1 a	24.3	y
Lake Champlain	294	2994.1±398.3	6.5±0.6	459.5	y
Arnold Bay	15	891.1±283.9 xyz	11.3±3.4 ab	79.1	y
Coast Guard	10	4.2±4.2 x	5.6±2.3 ab	0.8	n
Grand Isle	119	6584.5±875.7 xyz	10.6±1.0 b	618.3	y
Iron Bay	15	1.9±1.3 x	2.8±1.5 ab	0.7	n
Saxton Cove	15	108.9±58.3 xy	1.9±1.3 ab	58.0	y
Whallon Bay	120	680.4±142.7 xyz	2.5±0.5 a	276.0	y
Lake Michigan	1380	7.5±1.3	15.5±0.4	0.5	n
Nearshore	711	20.2±3.6	16.0±0.6	1.3	n
Bay Harbor	198	1.6±0.5 x	12.5±1.0 ac	—	n
Fishermans Island	182	2.9±0.8 x	7.8±0.8 bd	0.1	n
LTB-Crib	123	75.9±13.6	21.0±1.4 acd	3.6	n
GTB-Ingalls Point	160	0.4±0.2 x	25.6±1.5 a	0.01	n
Leeland	48	0.0±0.0 x	15.6±2.3 bcd	—	n
Offshore	669	0.2±0.1	14.4±0.6	0.01	n
Hog Island	169	0.0±0.0 x	10.5±1.0 bcd	—	n
Gull Island Shoal	160	0.8±0.3 x	27.4±1.4 a	0.03	n
Dahlia Shoal	164	0.0±0.0 x	8.8±0.9 bd	—	n
Middle Ground	117	0.0±0.0 x	12.5±1.2 bcd	—	n
Richards Reef	59	0.0±0.0 x	9.5±1.3 bcd	—	n

Note: In the last column, “y” indicates that fry were collected and “n” indicates no fry collected. Significant differences in the number of eggs per square metre and predators per square metre among sites within each of the lakes are indicated by different letters within a category and lake ($\alpha = 0.05$; Kruskal–Wallis multiple comparisons). SE, standard error; —, no data; LTB, Little Traverse Bay crib site; GTB, Grand Traverse Bay.

0.4 m⁻²) were more abundant than crayfish in Lake Champlain (1.0 ± 0.2 m⁻², K–W, $P < 0.001$), accounting for 68% of the species collected in egg bags. Similar to observations from Parry Sound, predator abundance levels in Lake Champlain were generally low, ranging from 2 to 11 m⁻² (average = 6.5 ± 0.6 predators·m⁻²). Sculpin were the most abundant predator in Lake Champlain (4.9 ± 0.4 m⁻²), yet densities were low and similar to those observed in Lake Michigan (5.5 ± 0.2 m⁻²; K–W, $P > 0.05$). Densities of sculpin in Parry Sound were the lowest (0.7 ± 0.2 m⁻²; K–W, $P < 0.05$; Fig. 2).

The length (size) of a predator can influence consumption rates. We examined differences in length between species and systems. *Orconectes propinquus* (20.7 ± 0.7 mm) from Parry Sound were larger on average than in Lake Michigan (15.0 ± 0.2 mm; K–W, $P < 0.001$). In Lake Michigan, *O. propinquus* (15.0 ± 0.2 mm) were smaller than *O. rusticus* (19.6 ± 0.3 mm; K–W, $P > 0.05$), and *O. rusticus* from Lake Michigan were similar in size to *O. propinquus* in Parry Sound (20.7 ± 0.7 mm; K–W, $P > 0.05$; Fig. 4). *Orconectes propinquus* were larger (22.7 ± 1.4 mm) when *O. rusticus* was present at a site than when it was absent (14.7 ± 0.2 mm; K–W, $P < 0.001$). *Cambarus robustus* (27.0 ± 1.7 mm), *O. limosus* (24.3 ± 1.5 mm), and *O. virilis*

(25.5 ± 0.6 mm) were the largest of the five species of crayfish observed (K–W, $P > 0.05$). However, the density and frequency of occurrence (< 2 m⁻²) of these species was substantially lower than that observed for either *O. rusticus* or *O. propinquus* (> 2 m⁻²; Figs. 3 and 5). In northern Lake Michigan, sculpin were longer (59.4 ± 0.5 mm) than round gobies (54.0 ± 2.6 mm; K–W, $P < 0.001$); this was likely because round goby populations were in the initial stages of invasion during this study and most individuals were young (J.L. Jonas, C.M. Diana, and R.M. Claramunt, unpublished data). The average size of sculpin was larger in Lake Michigan (62.1 ± 0.6 mm) than in Parry Sound (48.6 ± 4.2 mm) or Lake Champlain (49.7 ± 0.7 mm; K–W, $P < 0.05$).

Egg deposition

As expected, densities of eggs surviving to overwinter varied both within and among systems. Average egg densities in Lake Champlain (mean ± SE = 2994.1 ± 398.3 eggs·m⁻²) and Parry Sound (454.7 ± 36.3 ; K–W, $P > 0.05$) were significantly higher than those observed in Lake Michigan (7.5 ± 1.3 ; K–W, $P \leq 0.05$). Low levels of egg deposition occurred at only 5 of the 10 sites we evaluated in northern Lake Michigan, and extremely small numbers were detected at four of the sites (< 3 eggs·m⁻²; Table 1). A man-

Fig. 2. Proportion of species captured in egg bags in each of the three lakes, including four regions identified in Lake Michigan (Leeland, offshore, Grand Traverse Bay (GTB), and onshore). Native crayfish are considered egg predators and include *Orconectes propinquus*, *Orconectes virilis*, *Orconectes limosus*, and *Cambarus robustus*. Sculpin include mottled sculpin (*Cottus bairdi*) and slimy sculpin (*Cottus cognatus*). The “other” category primarily represents species that are not considered egg predators, including smallmouth bass (*Micropterus dolomieu*), largemouth bass (*Micropterus salmoides*), longnose dace (*Rhinichthys cataractae*), banded killifish (*Fundulus diaphanous*), johnny darters (*Etheostoma nigrum*), log perch (*Percina caprodes*), and bluntnose minnows (*Pimephales notatus*). Predators such as rock bass (*Ambloplites rupestris*), white sucker (*Catostomus commersoni*), longnose sucker (*Catostomus catostomus*), and mudpuppy (*Necturus maculosus*) were also included in the “other” category because they were infrequently observed.

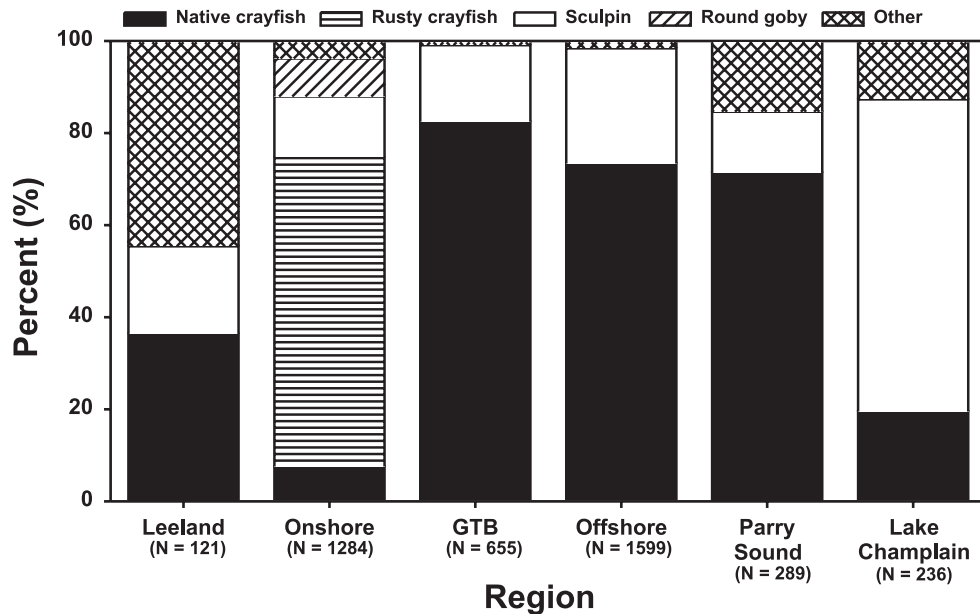
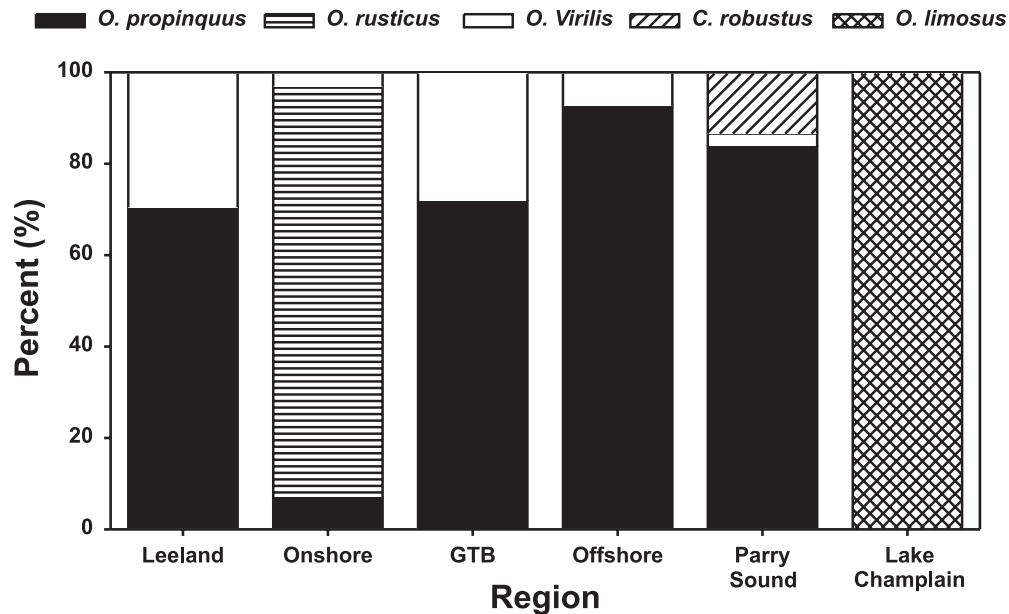


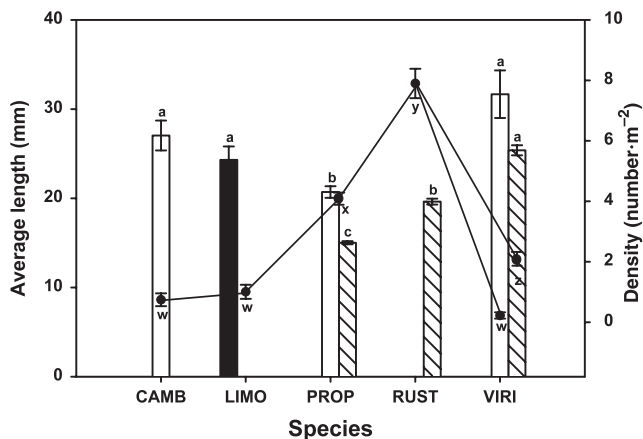
Fig. 3. The percent composition of crayfish species in Lake Champlain, Parry Sound, and four regions in northern Lake Michigan (Leeland, offshore, Grand Traverse Bay (GTB), and nearshore).



made crib located in northern Little Traverse Bay had the highest density of eggs (75.9 ± 13.6 eggs·m⁻²) when compared with other locations in Lake Michigan (K–W, $P < 0.001$). Lake trout spawned at all five sites in Parry Sound and egg density estimates were generally higher ($454.7 \pm$

36.3 eggs·m⁻²) than those observed in Lake Michigan (7.5 ± 1.3 eggs·m⁻²) and lower and less variable than in Lake Champlain (2994.1 ± 389.3 ; Table 1). Egg densities were highest at Bar and Davy islands (averaging 700–810 eggs·m⁻²; K–W, $P \leq 0.05$), intermediate at Horse and Mowatt (160 –

Fig. 4. Average density (number·m⁻², lines) and average length (mm, bars) summarized by region using sites where crayfish of a given species had established populations. Crayfish species are represented by four letter codes: CAMB, *Cambarus robustus* (Parry Sound sites only); LIMO, *Orconectes limosus* (Lake Champlain sites only); PROP, *Orconectes propinquus* (all Parry Sound and Lake Michigan sites); RUST, *Orconectes rusticus* (on-shore Lake Michigan locations only); and VIRI, *Orconectes virilis* (Lake Michigan and Parry Sound). Bars with the same letters are not significantly different within a variable. With the exception of *O. virilis*, the density of predators did not differ between regions within a species group (K–W, $P > 0.05$); therefore, average values are provided for all regions combined. Solid bars represent Lake Champlain, open bars represent Parry Sound, and hatched bars represent Lake Michigan.



215 eggs·m⁻²; K–W, $P \leq 0.05$), and lowest at Mary (30 ± 9 eggs·m⁻²; K–W, $P \leq 0.05$; Table 1). We measured the highest density of eggs in Lake Champlain at Grand Isle (6584.5 ± 875.7 eggs·m⁻²; K–W, $P \leq 0.05$); this location is an extreme outlier when compared with other locations. Egg deposition at other locations in Lake Champlain ranged from 2 to 891 eggs·m⁻² and was high overall in the lake, with a mean (±SE) of 552.7 ± 102.9 eggs·m⁻².

Egg-to-predator ratios

To establish a simple measure of the relationship between egg deposition and predator abundance as they relate to fry emergence in the spring, we estimated the number of lake trout eggs per predator in a square-metre area at each site. Estimates of eggs per predator were compared among sites with and without measured fry emergence (an indication of success in surviving over the winter). Lake Michigan had the lowest ratios (<4 eggs·predator⁻¹·m⁻²). The average number of eggs per predator in Parry Sound was nearly 20 times that of Lake Michigan and one sixth that of Lake Champlain at 82.1 eggs·predator⁻¹·m⁻². Lake Champlain had the greatest average egg-to-predator ratio (459.5 eggs·m⁻²), nearly 114 times that of Lake Michigan. If the high egg densities at Grand Isle (an apparent outlier) are removed from the analysis, the average ratio becomes similar to Parry Sound at 85 eggs·predator⁻¹·m⁻², still nearly 21 times that of Lake Michigan. Fry emergence was not detected at any of the Lake Michigan sites; however, fry emergence was detected at four of five sites evaluated in Parry Sound, and four of six sites evaluated in Lake Champlain, an indication of success

in surviving the overwinter period (Table 1; Fig. 5). The lowest number of eggs measured per predator at a successful site was 24 eggs·predator⁻¹·m⁻² at Mowatt Island in Parry Sound. Successful lake trout populations resulting in fry emergence are likely produced when egg densities are high enough to survive initial predation pressure. In our evaluation, this was generally observed when there were more than 80 eggs·predator⁻¹·m⁻². Average ratios across successful sites reveal that egg densities exceeding 190 eggs·predator⁻¹·m⁻² one month post-spawn will likely succeed in producing emergent fry.

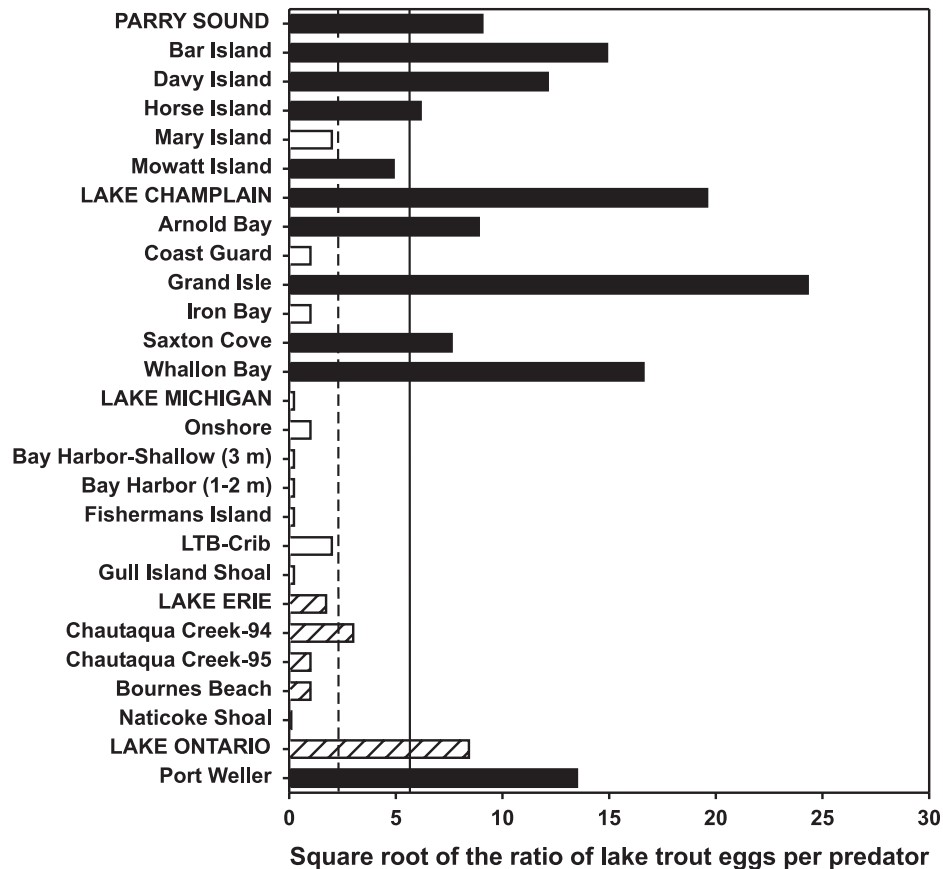
Discussion

Management strategies for lake trout have primarily focused on estimates of adult abundance. We observed significant differences in the number, type, and size of predators, as well as in the number of eggs surviving to overwinter among the systems and sites that we evaluated. Given these differences, we suggest a management strategy that is related to the current strategy, but that shifts the focus to measures of surviving eggs in place of the relative abundance of adults. Interstitial predator communities in northern Lake Michigan were the most lethal. Larger, more abundant exotic species were present at most nearshore locations, while off-shore and Grand Traverse Bay communities were similar to those in Parry Sound and were dominated by native crayfish (mainly *O. propinquus*). Total predator densities in Lake Michigan were nearly twice those observed in Lake Champlain or Parry Sound. Sculpin were not only larger in Lake Michigan, but were more abundant when compared with populations in Parry Sound and equal to those observed in Lake Champlain. The second highest potential for predation occurred in Lake Champlain, where overall predator densities were similar to those in Parry Sound, but sculpin were the predominant predator.

Egg deposition was high in Parry Sound and Lake Champlain but barely detectable in Lake Michigan. Egg abundance levels in northern Lake Michigan are among the lowest in the Great Lakes assessed using similar types of gear (Marsden 1994; Fitzsimons 1995a; Fitzsimons and Williston 2000), and likely have not increased substantially since early lake trout rehabilitation efforts in the 1970s and early 1980s. Low egg-deposition rates observed in northern Lake Michigan are most likely attributable to a combination of low spawner abundance levels (Bronte et al. 2003a) and high interstitial predation rates within the first month after the peak spawning period.

The metric used by most managers to build successful adult stocks (50 spawning lake trout·305 m⁻¹ of gillnet; Selgeby et al. 1995), may not represent a suitable target stock size for Lake Michigan, given high levels of interstitial predation. Additionally, stocked lake trout are likely less efficient spawners than wild fish because selective pressures favor hatchery rearing over production in the lake. When data for the current metric were collected (1973–1992), restoration efforts had only been accomplished in Lake Superior, where self-sustaining lake trout stocks had been re-established. Despite reports of adequate adult abundances at many sites in Lake Michigan, there have been no reports of measurable recruitment. We can infer from the lack of mea-

Fig. 5. Ratio of lake trout (*Salvelinus namaycush*) eggs deposited-predator⁻¹·m⁻² at sites in northern Lake Michigan, Lake Champlain, and Parry Sound. Data from Lake Erie and Lake Ontario collected in previous investigations (J.D. Fitzsimons, personal communication) are also included. The broken line represents the largest ratio where fry emergence was not detected and the solid line represents the lowest ratio at sites with detectable fry emergence. Bars next to lake names represent the square root of average ratios for the region. To better represent small numbers, data were transformed to the square root of values. Solid bars indicate sites where emergence was detected, open bars indicate sites with no emergence, and hatched bars indicate sites where fry emergence was not assessed. LTB-Crib, Little Traverse Bay crib site.



surable recruitment that either there was insufficient spawning activity, high egg mortality, or some combination of the two. Lake Michigan may require higher numbers of spawning adults to recruit future year classes of lake trout.

Improved understanding of spawning habitat use and behaviors should prove to be useful to managers wishing to establish target egg densities. Our data indicate that, even in the face of abundant interstitial predators, it is possible to swamp a given area with enough eggs to allow for survival to the fry stage. Depensation likely occurs as more eggs are deposited per unit area and the available predators are satiated. Most predation occurs within the first 28 days after eggs are deposited, and eggs surviving this period are highly likely to survive through the winter period (Dittman et al. 1998; Claramunt et al. 2005). By concentrating spawning activity on certain areas, it may be easier to achieve target egg densities rather than focusing on regional or lake-wide abundance targets for adults.

Ratios of eggs per predator provide a simplistic way to index the effectiveness of lake trout rehabilitation efforts. In this case, the target density of eggs per predator is based on initial egg deposition minus the effect of predation and other losses prior to entering the winter period. The lowest num-

ber of eggs deposited per predator at locations that were successful in producing viable lake trout fry in our evaluations was 24 eggs-predator⁻¹·m⁻². Figure 5 represents a summary similar to that presented in Selgeby et al. (1995), using ratios instead of abundance to indicate reproductive potential. The average among the 10 successful spawning locations evaluated in this investigation was 190 eggs-predator⁻¹·m⁻²; this may be a more reliable target for managers to ensure survival to the fry stage given additional density-independent sources of mortality and variation among systems. In Lake Michigan, where the highest egg-to-predator ratio was 4 eggs-predator⁻¹·m⁻², egg deposition would need to increase 20-fold to obtain egg survival similar to the average in Parry Sound (82 eggs-predator⁻¹·m⁻²). Data from other investigations support these recommendations. Peck (1986) found egg-to-predator ratios range from 61–259 eggs-predator⁻¹·m⁻² at sites with successful fry emergence in Presque Isle (Lake Superior). Work on Keuka Lake, New York, where lake trout are self-sustaining, has yielded ratios of 192 eggs-predator⁻¹·m⁻² (Fitzsimons et al. 2005).

More detailed analyses of species assemblages and the size and density of predators at spawning sites provides further insight into the overall potential for predation effects

(Horns and Magnuson 1981). The size of a predator directly influences the number of eggs consumed, while abundance levels, competition, and foraging behaviors likely strengthen or weaken this relationship in both conspecific and mixed-species environments (Gill and Hart 1996; Biga et al. 1998; Anderson 2001). Based on the potential for consumption of eggs, the hierarchy of predators observed in our study is fish > crayfish; round gobies > sculpin; and rusty crayfish > native crayfish. Exotic species in the latter two comparisons are more effective predators as they tend to be larger in size and tolerate dense populations of conspecifics.

Because crayfish dominated many of our spawning locations and fish dominated others, it is important for us to understand differences in foraging efficiency between fish and crayfish. The potential vulnerability of eggs may vary based on the type of predator and their physical position within rocky substrates. In studies comparing sculpin with *O. virilis*, predation rates were similar for the two species independently, but fish had a negative impact on crayfish foraging efficiency when species were together (Miller et al. 1992; J.D. Fitzsimons, G. Fodor, and B. Williston, unpublished data). Sculpin ate more eggs than *O. propinquus* in laboratory studies; however, when combined, crayfish negatively influenced sculpin populations by denying them access to refuge areas, indirectly increasing their vulnerability to predation (McNeely et al. 1990; Miller et al. 1992). In summary, crayfish were less efficient predators than sculpin, but likely influence the abundance of benthic fish populations through competition for habitat and are less vulnerable to predation (Olsen et al. 1991).

Crayfish were the most abundant predators in Parry Sound and Lake Michigan. *Orconectes rusticus* dominated the nearshore zone of northern Lake Michigan and *O. propinquus* dominated other locations in Lake Michigan and Parry Sound. Our work compliments other investigations by showing that, when both were present, the average size of *O. rusticus* was larger than that of *O. propinquus*, and that densities of *O. rusticus* were nearly twice as high (Olsen et al. 1991). *Cambarus robustus*, *O. limosis* and *O. virilis* were the largest of the five species of crayfish observed; however, the density and frequency of occurrence of these species was the lowest. Of the two predominant species of crayfish (*O. rusticus* and *O. propinquus*), *O. rusticus* would appear to be the more formidable predator; its higher density and larger size lead to higher potential consumption levels per individual when compared with the smaller *O. propinquus* and other large but less common species of native crayfish. Further, *O. rusticus* appears to be displacing *O. propinquus* when present, so populations are likely to increase in nearshore habitats throughout the Great Lakes.

Sculpin were the most abundant predator in Lake Champlain, and were the most common of the fish species collected in Parry Sound and Lake Michigan. In our investigation, the abundance of sculpin was lowest at sites with round goby populations when compared with sites without round gobies, suggesting that competitive displacement may be occurring since their arrival in northern Lake Michigan (Janssen and Jude 2001). In other investigations, round gobies have been found to interfere with breeding activities and compete for food and resources with sculpin

(Dubs and Corkum 1996; Janssen and Jude 2001). In laboratory studies, round gobies were significantly more efficient at finding lake trout eggs at low densities ($63 \text{ eggs} \cdot \text{m}^{-2}$) than sculpin, and were found to eat three times the number of eggs (J.D. Fitzsimons, G. Fodor, and B. Williston, unpublished data). Round goby populations continue to increase (J.L. Jonas, C.M. Diana, and R.M. Claramunt, unpublished data), and like rusty crayfish, further growth and expansion of goby populations is likely. Based on their tendency toward higher population levels and greater individual search and egg consumption rates, round gobies are likely superior predators when compared with sculpins.

The abundance of mature lake trout occupying a given spawning location likely relates to the density of viable eggs deposited. To compare the relative abundance of adults among the three lakes, we have evaluated published accounts from other agencies (Reid et al. 2001; Bronte et al. 2003b; Ellrott and Marsden 2004). Estimates of spawner abundance on sites in northern Lake Michigan were extremely low. In recent multi-agency fall assessment efforts, non-stocked offshore locations produced fewer than $25 \text{ fish} \cdot 12 \text{ h}^{-1} \cdot 305 \text{ m}^{-1}$ gillnet, and nearshore locations produced between 35 and 50 lake trout $\cdot 12 \text{ h}^{-1} \cdot 305 \text{ m}^{-1}$ gillnet (Bronte et al. 2003a). Estimates from Parry Sound, after adjusting catch per unit effort from lake trout $\cdot 1.5 \text{ h}^{-1} \cdot 91.7 \text{ m}^{-1}$ gillnet to lake trout $\cdot 12 \text{ h}^{-1} \cdot 305 \text{ m}^{-1}$ gillnet, were significantly higher than those observed for Lake Michigan, and ranged from 88 to 114 lake trout $\cdot 12 \text{ h}^{-1} \cdot 305 \text{ m}^{-1}$ gillnet (Reid et al. 2001). Fall lake trout abundance estimates in Lake Champlain are not directly comparable with the other lakes, as lake trout were only assessed using electroshocking. In Lake Champlain, high egg deposition rates coincide with high spawner abundance levels primarily a result of stocking. Nearly twice the number of lake trout has been stocked per unit area in Lake Champlain ($127 \text{ yearling equivalents} \cdot \text{km}^{-2}$) when compared with Lake Michigan ($72 \text{ yearling equivalents} \cdot \text{km}^{-2}$). Parry Sound populations had been supplemented with stocked fish until 1997. Management actions increasing populations of spawning lake trout would likely elevate the number of eggs deposited per predator in a unit area.

The results of our investigation could be used to establish target densities of eggs deposited at a given location, and with further work, the target concentration of spawning adults needed to accomplish this. It is important to note that loss at the egg stage represents the first potential bottleneck to lake trout recruitment. We have not evaluated the influence of predation on post-swim-up fry, potentially the second bottleneck to lake trout recruitment.

Ecologists have hypothesized that predation may be the most important source of egg and larval mortality for fishes (Houde 2002). The mortality of early life stages due to predation represents a density-dependent response and is likely to decrease with increases in the numbers of lake trout eggs deposited and increase with higher abundances of predators. Other mortality factors may be density independent (i.e., thiamine deficiencies and physical disturbances) and can be responsible for significant losses at the egg and fry stage (Perkins and Krueger 1995; Fitzsimons et al. 1999). Density-dependent and independent sources of mortality should be considered in concert when determining the potential for

survival of lake trout eggs and fry, because stock recruitment relationships require a measure of the reproductive potential of the population and the number of young that will reach maturity.

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